

ITA05-ACTIVITY

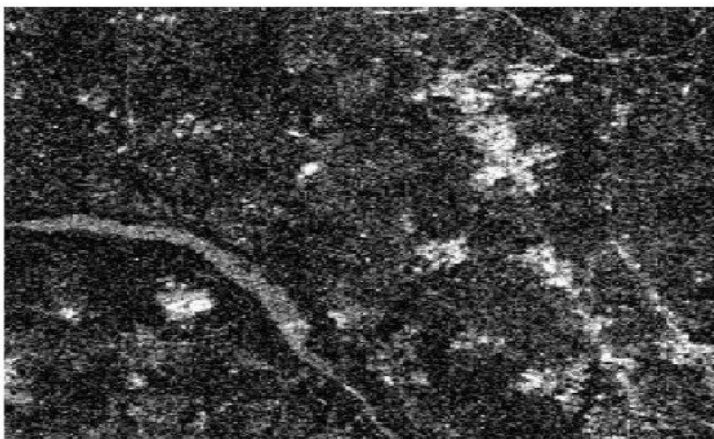
SCENARIO BASED ACTIVITY

Mention the techniques used with reason:

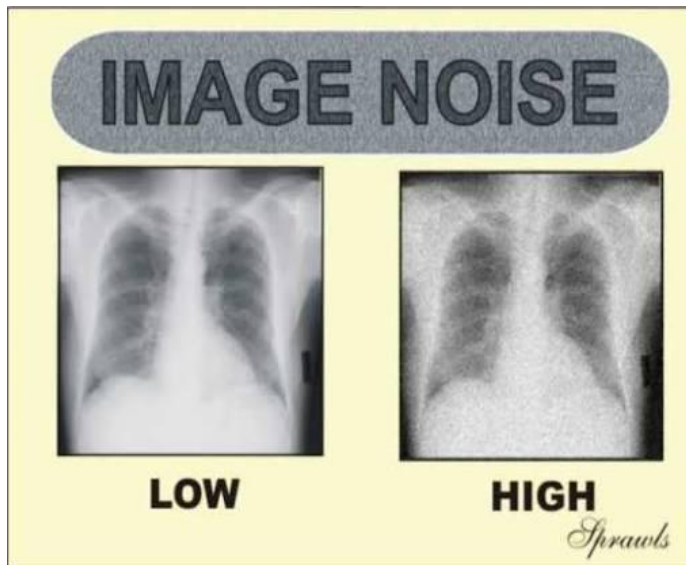
1. A CCTV image contains **salt-and-pepper noise** due to transmission errors.



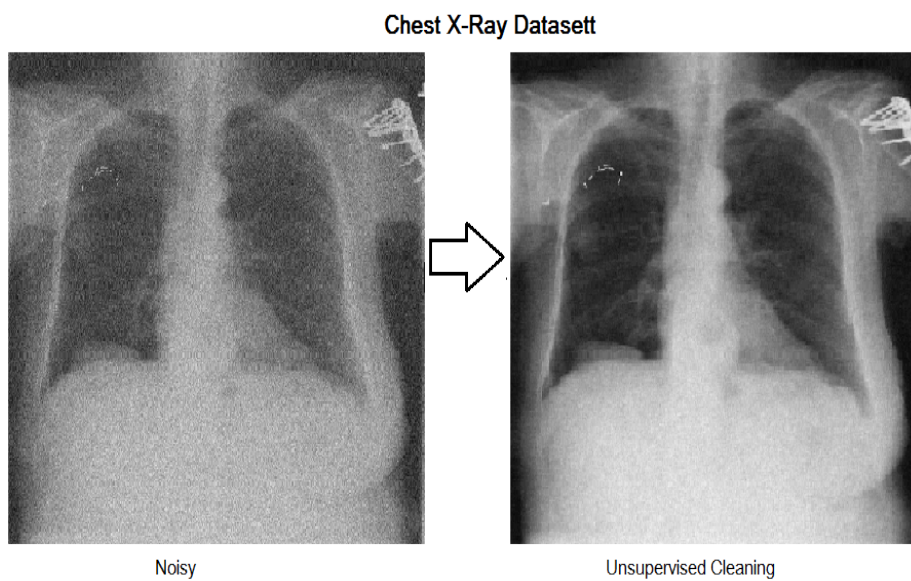
2. A satellite image is corrupted by **Gaussian noise** from sensors.



3. A medical X-ray image has random noise but fine edges must be preserved.

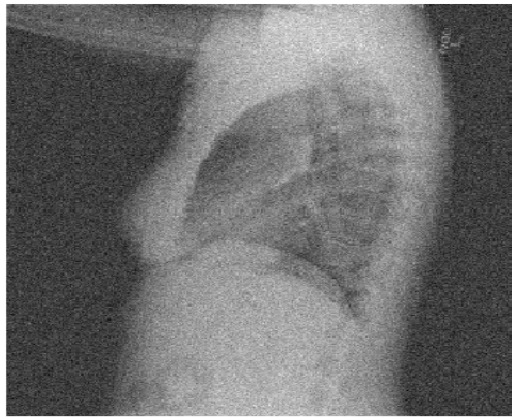


4. A scanned document has **black and white dots** scattered randomly.

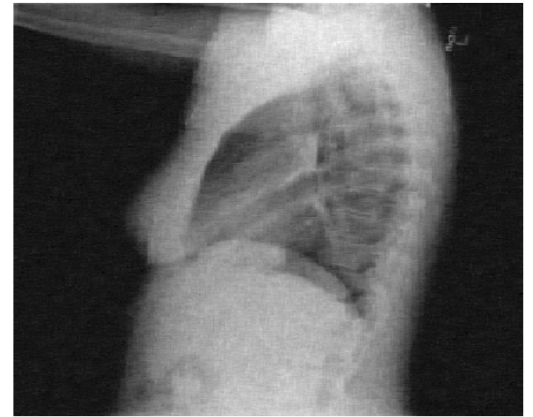
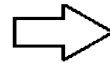


5. A low-light image shows heavy grain without clear boundaries.

Indiana University X-Ray Dataset



Noisy



Unsupervised Cleaning

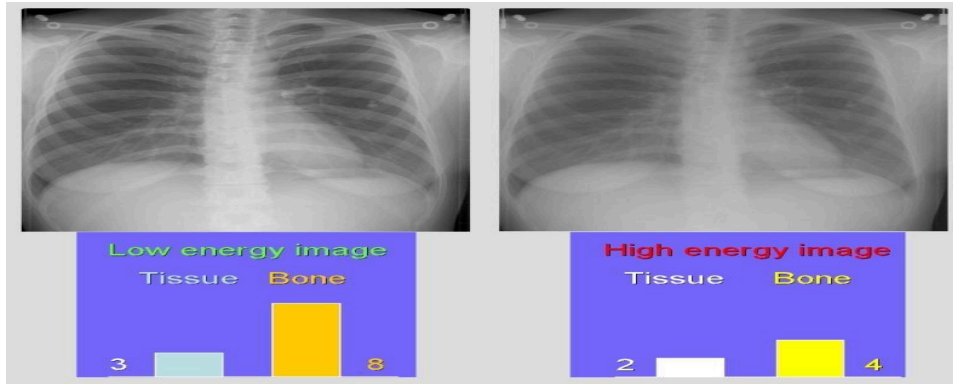
6. A foggy road image has **very low contrast**.



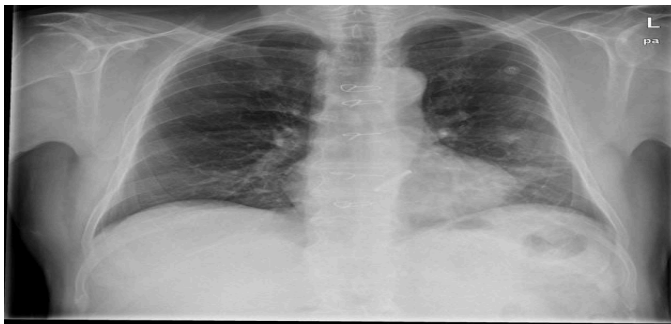
7. An underwater image appears dull and washed out.



8. A chest X-ray shows poor visibility of internal structures.



9. A night-time traffic image lacks intensity variation.



10. A grayscale image uses only a narrow intensity range.



Activity-2

- 1) Median Filter:- It is used to remove salt and pepper noise from CCTV image. It preserves image details while eliminating noisy pixels.
- 2) Gaussian filter:- It is used to reduce Gaussian noise in satellite images. It smooths the image and minimizes sensor noise.
3. Bilateral filter:- It removes random noise while preserving fine edges. It is ideal for medical x-ray images.
4. Median filter:- It removes randomly scattered black and white dots in scanned documents. It improves document clarity.
5. Gaussian filter:- It reduces heavy grain in low-light images. It smooths the image and improves visibility.
6. Histogram Equalization:- It enhances the contrast of foggy road images. This makes objects and roads more visible.
7. Contrast stretching:- It improves the brightness and contrast of underwater images. It makes the image appear clearer.

8). CLAHE: It enhances local contrast in chest X-ray image. It improves visibility of internal structure without amplifying noise.

9. Histogram Equalization: It increases intensity variation in night time traffic image. This improves overall image visibility.

10. Contrast stretching: It expands the narrow intensity range of grayscale image. This enhances the overall image contrast.

